Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

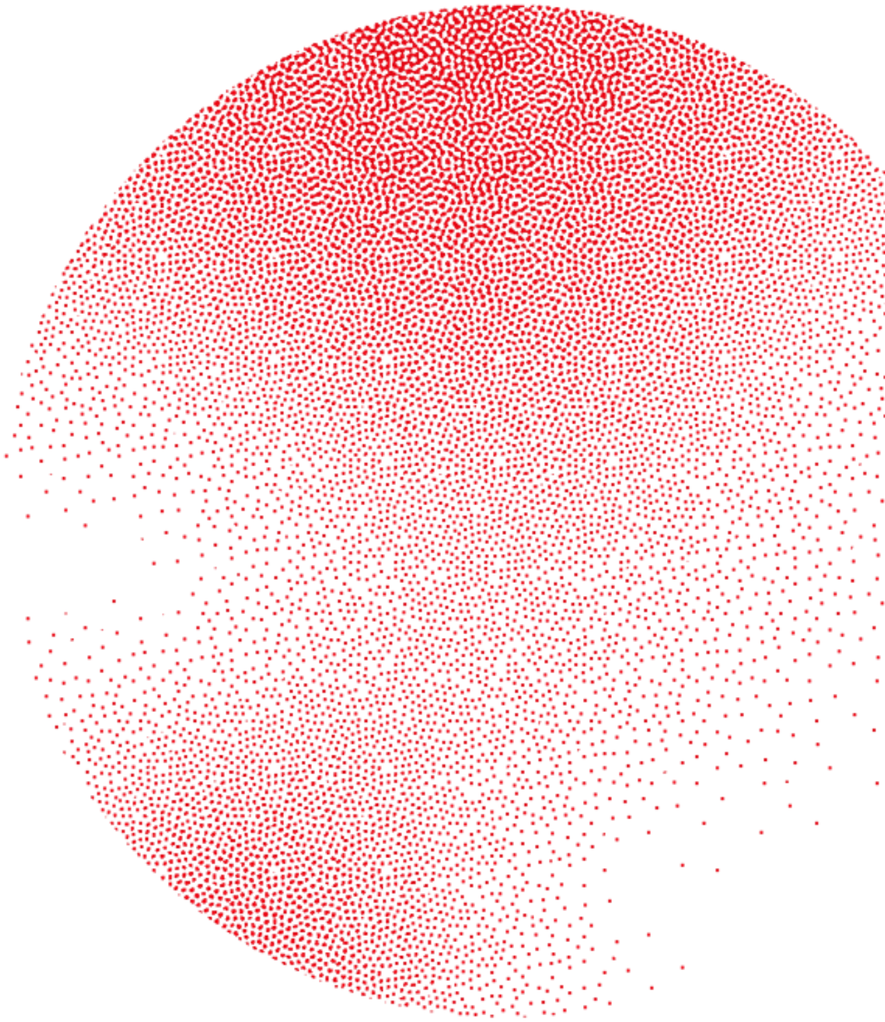
Day 1 - Session 2

PhD Jeferyd Yepes-García
University of Fribourg
October 26th, 2026

elixir SWITZERLAND

This presentation

- Learning outcomes
- Sequencing methods
- Sequencing by synthesis
- Long-read technologies
- Data cleaning and quality control
- Summary



Learning outcomes

- Recognize the main sequencing technologies and how the choice of short- or long-read platforms influences the experimental design and the analysis strategy.
- Explain why quality control (QC) is a critical step in any metagenomics workflow.
- Interpret the main modules of a FastQC report and compare raw vs cleaned reads.

Reminder of the sequencing methods



1986 - 2001 | 2005 | 2006 | 2010 | 2011 | 2014 | 2018

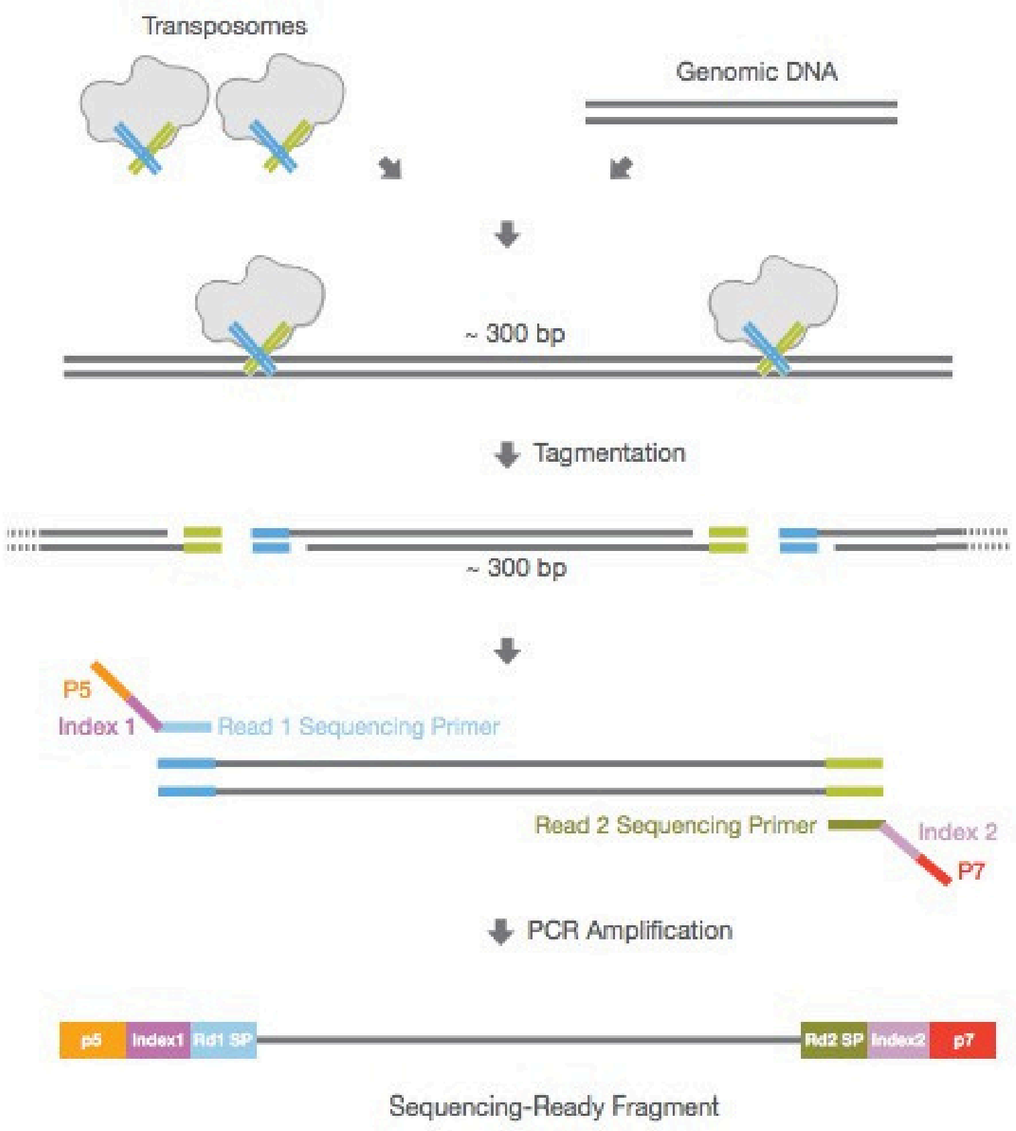
ABI Sanger | Roche 454 | Illumina | Life SOLiD | Life Ion Torrent | Pacific Biosciences | Oxford Nanopore

First Generation

Second Generation

Third Generation

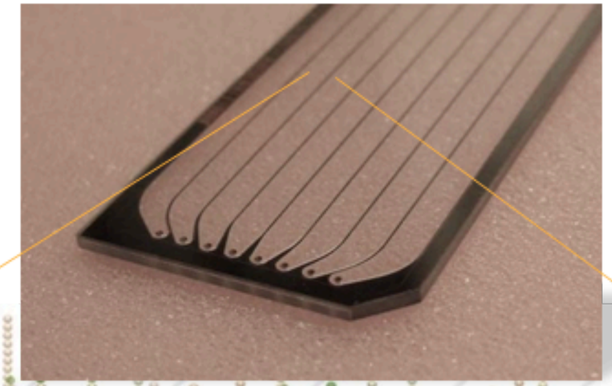
The most used sequencing method: Sequencing by synthesis

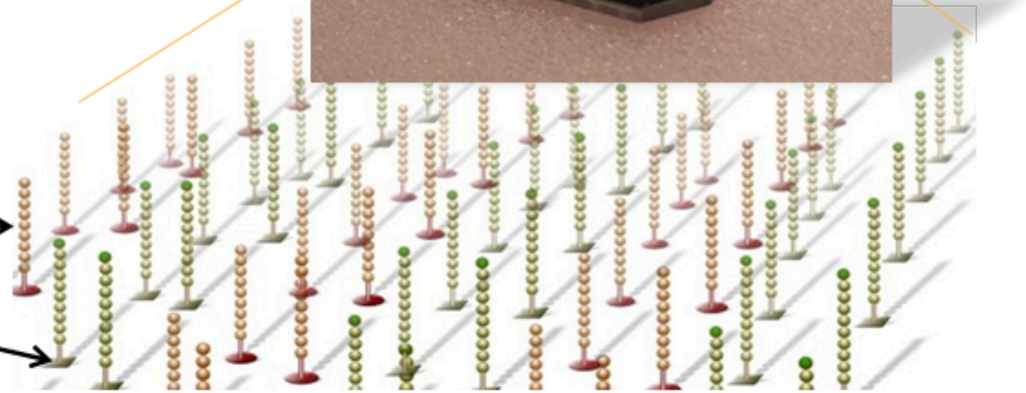


The most used sequencing method: Sequencing by synthesis

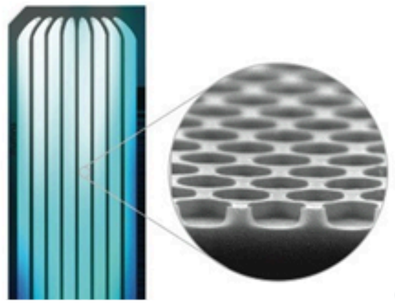
A flow cell is a thick glass slide with several channels or lanes

Each lane is randomly coated with a lawn of oligos that are complementary to library adapters or patterned



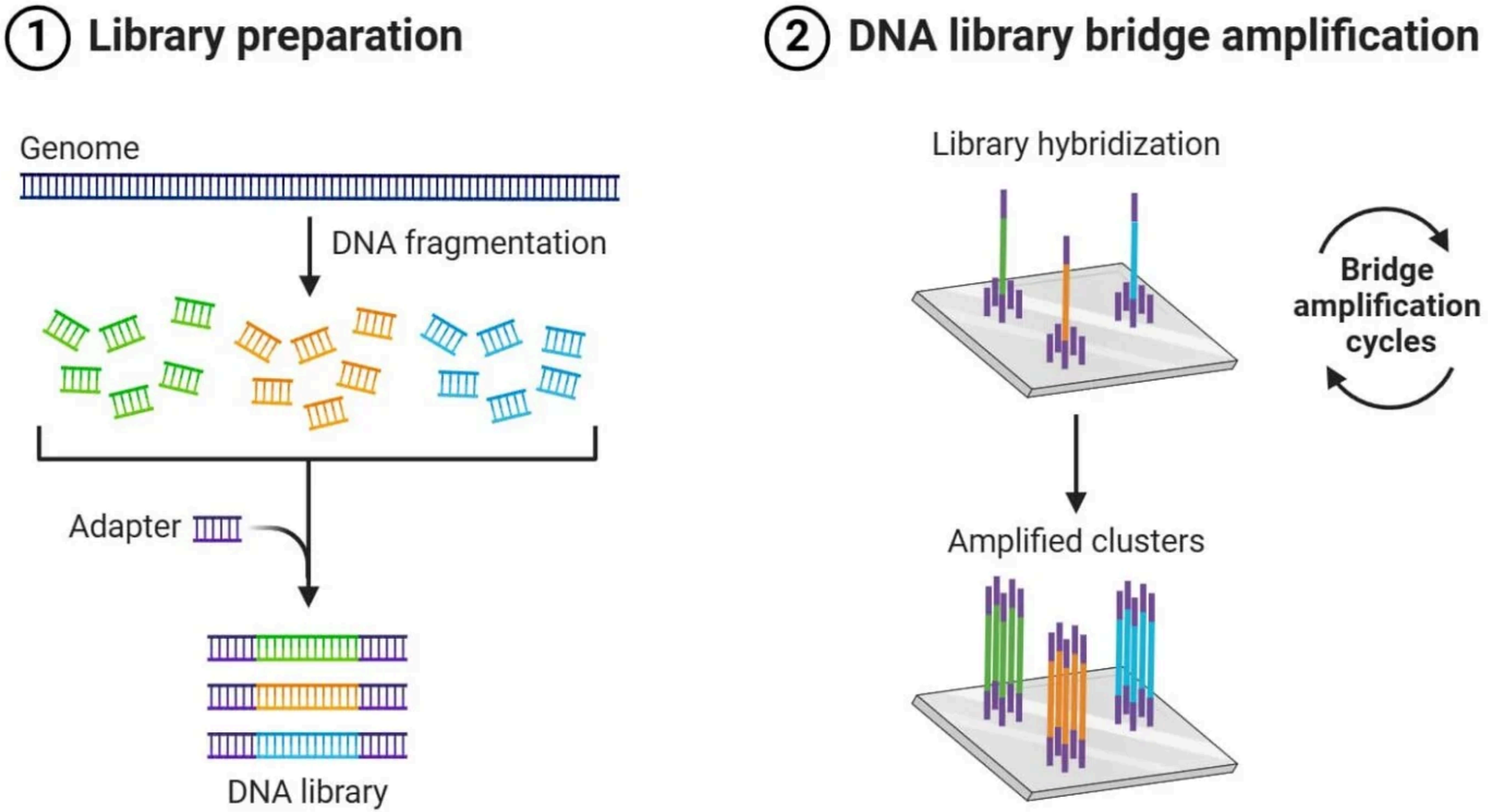






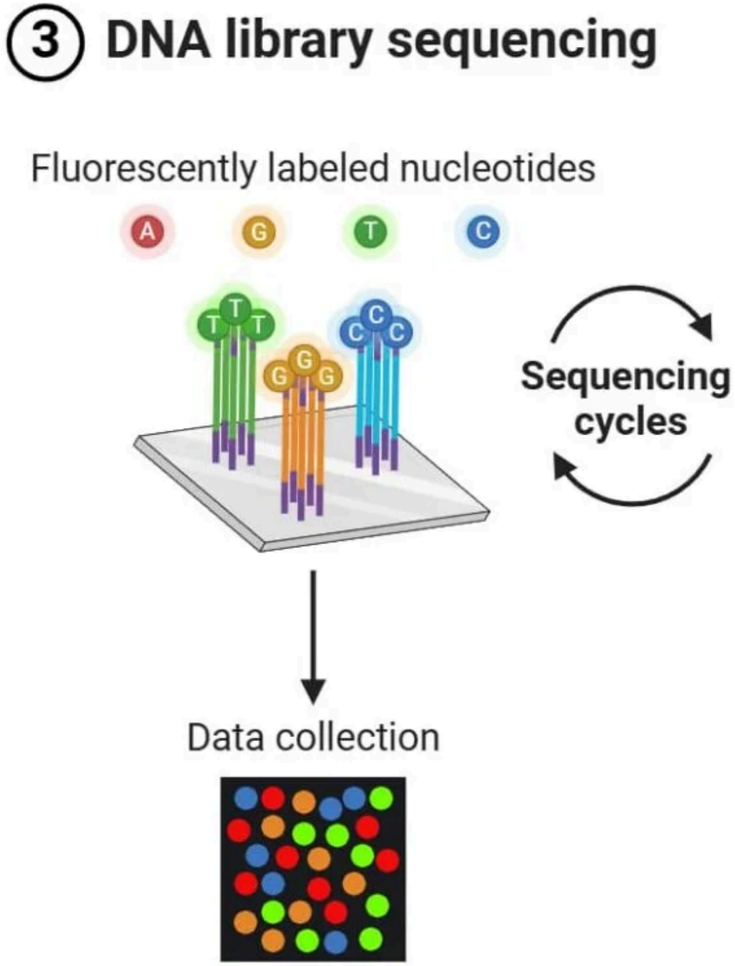
Patterned Flow Cells: higher number of clusters

The most used sequencing method: Sequencing by synthesis



Tamang (2024)

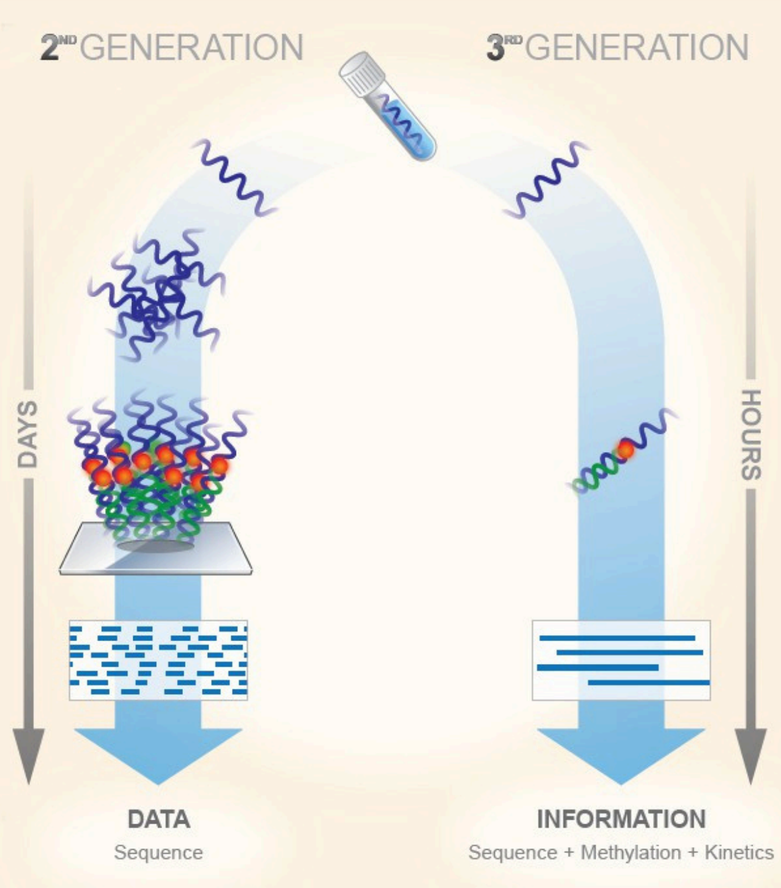
The most used sequencing method: Sequencing by synthesis

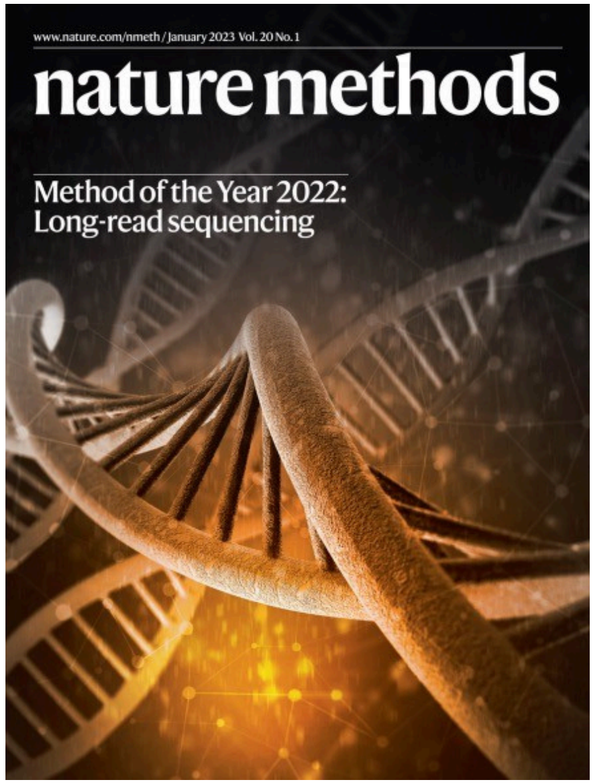


Tamang (2024)

The Third Generation Sequencing Platforms: PacBio and Oxford Nanopore

Lower throughput, but longer reads and additional information (base modifications)



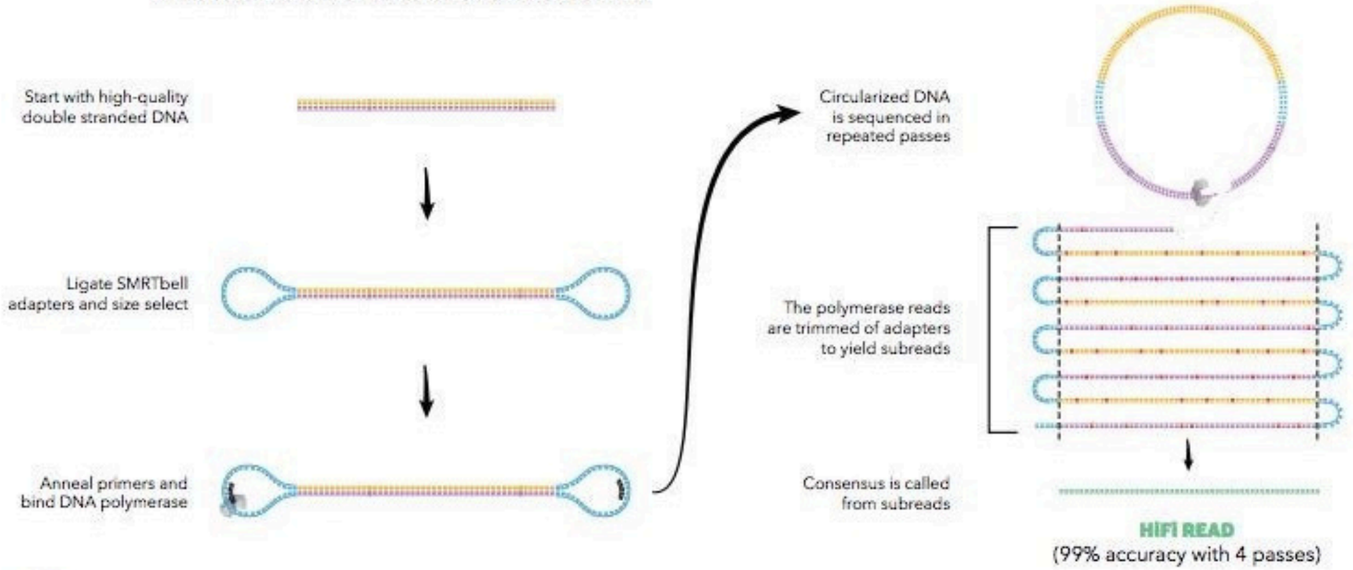


HiFi Reads

HIFI READS

Generate Highly Accurate Long Reads

Produce HiFi reads using the circular consensus sequencing (CCS) mode to provide base-level resolution for detection of all variant types from single nucleotide to structural variants.



LONG READS

Optimize Your Run for Even Longer Reads

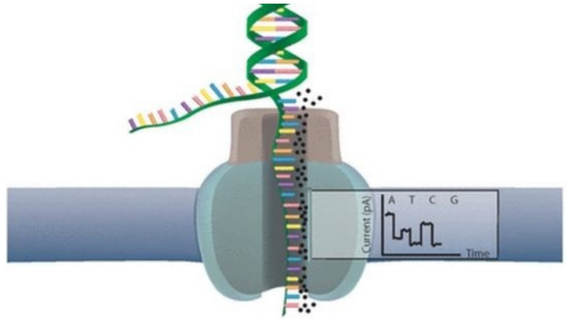
Sequence read lengths in the tens of kilobases using the continuous long read (CLR) sequencing mode to enable high-quality assembly of even the most complex genomes.

Half of Data in Reads
>50 kb

Longest Reads Up To
175 kb

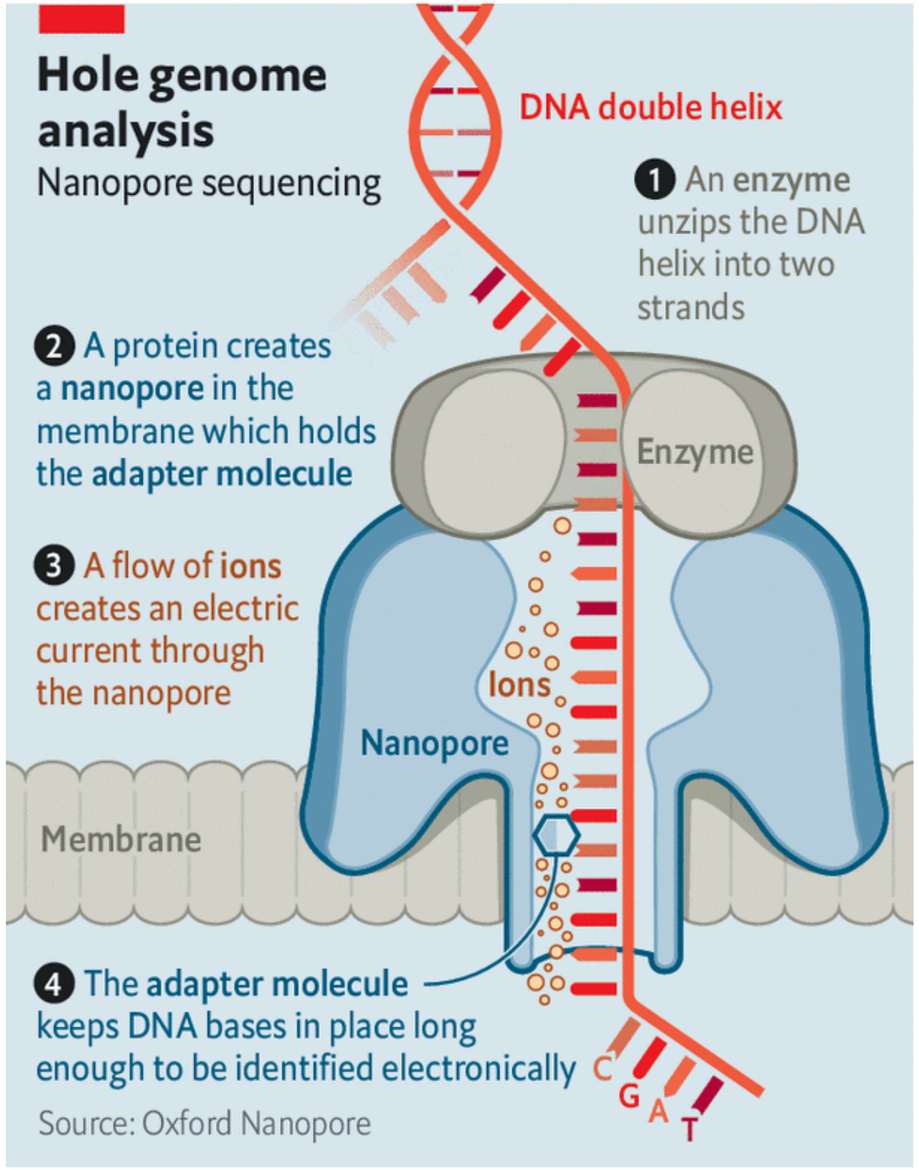
Oxford Nanopore





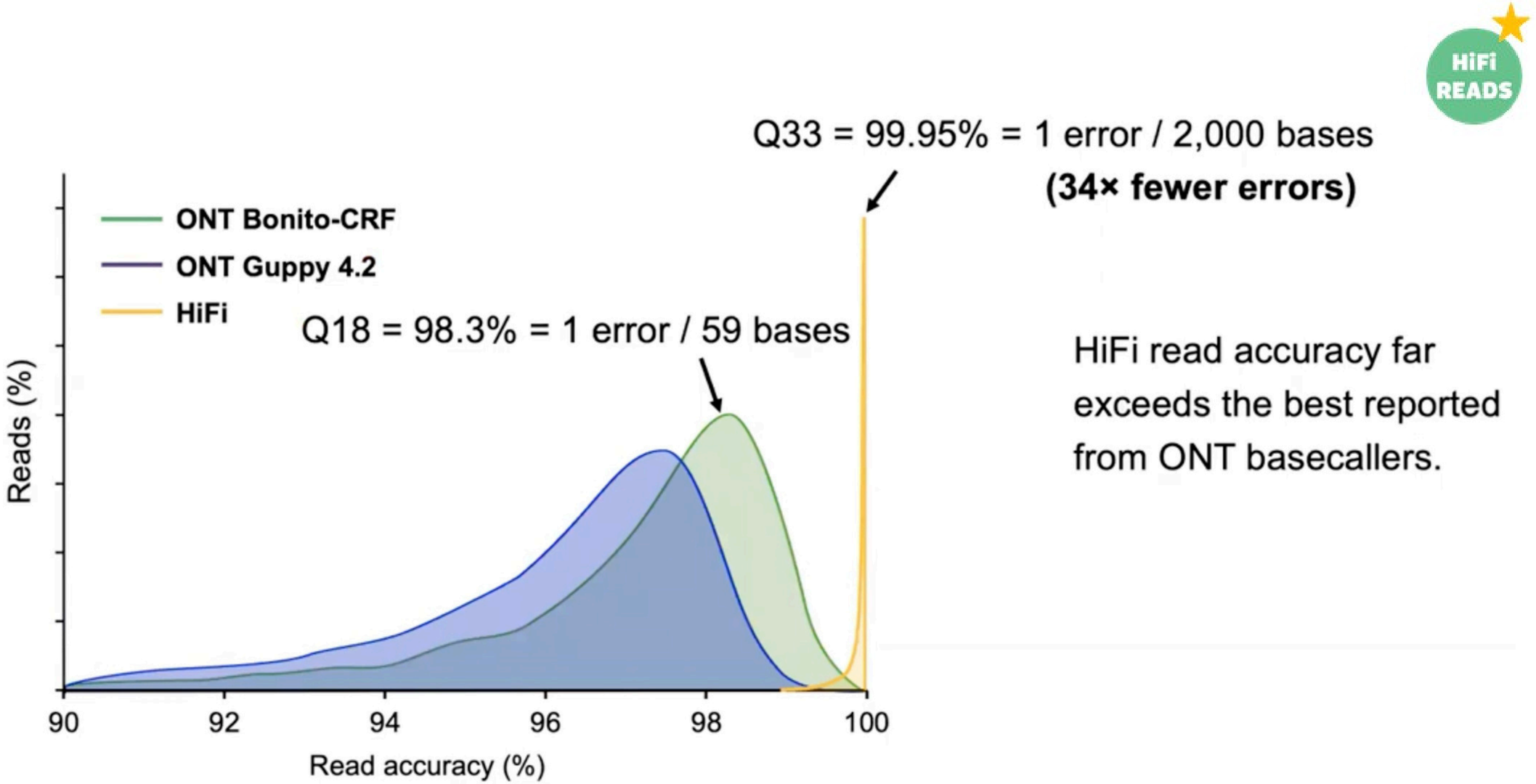
Error rate still high!
indels 13%
substitution 5%

Recently improved
<2% with R10.4



The Economist

HiFi reads compared to ONT



HiFi read accuracy far exceeds the best reported from ONT basecallers.

HiFi: HG003 18 kb library, Sequel II System Chemistry 2.0, precisionFDA Truth Challenge V2
ONT: Bonito-CRF & Guppy 4.2 NCM Nanopore Tech Update Dec. 2020

Short reads vs Long reads

Short reads	Long reads
Illumina or MGI	**PacBio (HiFi) or Nanopore**
Highly flexible usage	High quality genome assembly
	DNA base modifications detection
High quality reads of identical length	High quality HiFi reads (PacBio)
Enormous throughput, low price per base	Cheap apparatus (Nanopore)
Limitations	**Limitations**
Short (max 300bp PE)	Uneven length
Not suitable to solve repeats	High error rate in raw reads (mainly indels)

Advantages of long reads

High quality genome assembly

- Structural variation (such as inversions or insertions)
- Complex rearrangements
- Repeat expansion length variations
- Variants in repetitive or highly polymorphic regions
- Telomere-2-telomere
- Haplotigs separation or phasing

Other areas

- DNA base modifications detection
- Transcripts isoforms identification



Data cleaning & Quality control

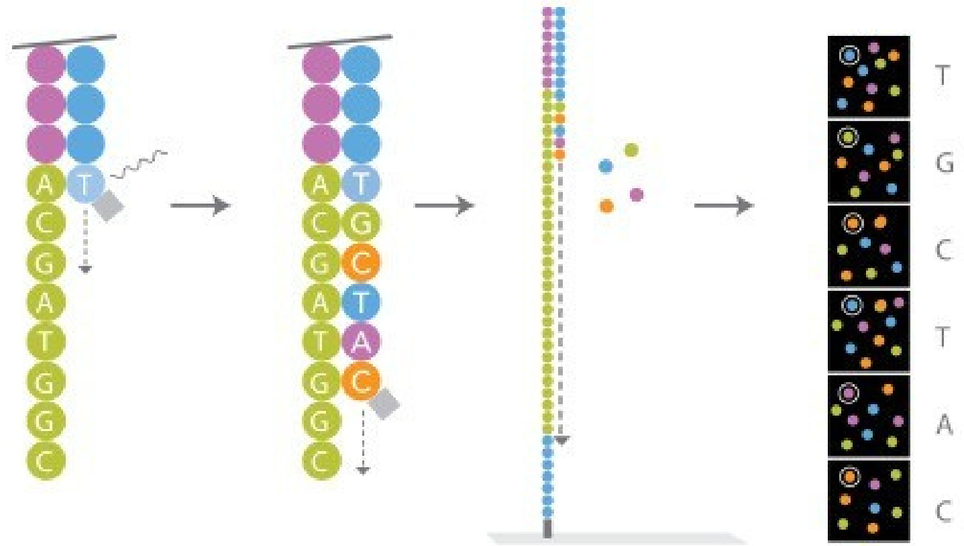


What do we receive from the sequencing facility?

The sequenced part of one DNA fragment.

These DNA sequences are given by the sequencing machine with a

Phred quality score in so called FASTQ format




```
@SRR001666.1  071112_SLXA-EAS1_s_7:5:1:817:345  length=36
TGCTACTGGCCGCTGCCGATGGCGTCAAATCCCACC
+SRR001666.1  071112_SLXA-EAS1_s_7:5:1:817:345  length=36
IIIIIIIIIIIIIIIIIIIIIIIIIIIII9IG9IC
```

Phred quality score, a measure of base call quality



$$Q\text{sanger} = -10 \log_{10} p$$

Phred quality scores are logarithmically linked to error probabilities

Phred Quality Score	Probability of incorrect call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

The quality score is ASCII encoded in the FASTQ format

FASTA vs FASTQ

```
>C3Po_0001:2:1:17:1499#0/1
TGAATTCATTGACCATAACAATCATATGCATGATGCAAATTATAATATCATTT
TTGTTTGAGCAAATGATTCATAATAATGTATTTCAATATTTTTAGGAATATCT
CCCAATATTGCGCGTGCTGAATTCCATCCGGAATTTTTGACGTCCCCCCCCGA
ANGGANGNGANNNNGNNGNNNTNTNNAAANGNNNNN
…
```

```
@M01867:115:000000000-ABF5V:1:1101:9268:1666  1:N:0:51
AACAGGATTAGATACCCTGGTAGTCCACGCCCTAAACGATGCGAACTGGTTGTTGGGTGCTTTTTG
+
--A-6@8CE,@<CEFGGFAFF9CEFF,C@CE@B<8@C:CC,,+,7@C<6,668C,,+8,6,,<9,+
@M01867:115:000000000-ABF5V:1:1101:9214:1685  1:N:0:51
AACCGGATTAGATACCCTGGTAGTCCACGCCCTAAACGATGTCTACTAGTTGTTGGTGGAGTAAAA
+
--AA@7:FF9C9C@FEFE<CF9FEFF,C@FE:B8,6C:+C6CFD9CE,<C6<C@,,8,,,,;,,,-
@M01867:115:000000000-ABF5V:1:1101:18344:1708  1:N:0:51
AACAGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCAACTAGCCGTTGGGAGCCTTGAG
+
--A99E8CE<C9CFFGGG8FF9@CFF9ECFF@F;,CFC7C,CF,,CF@@EE@@@,,+,,6BE@,,-
…
```

read 1

read 2

read 3

FASTQ is a FASTA with score!

Quality Control of the data

First step after receiving the data
Sometimes already done by the sequencing center

Objective:

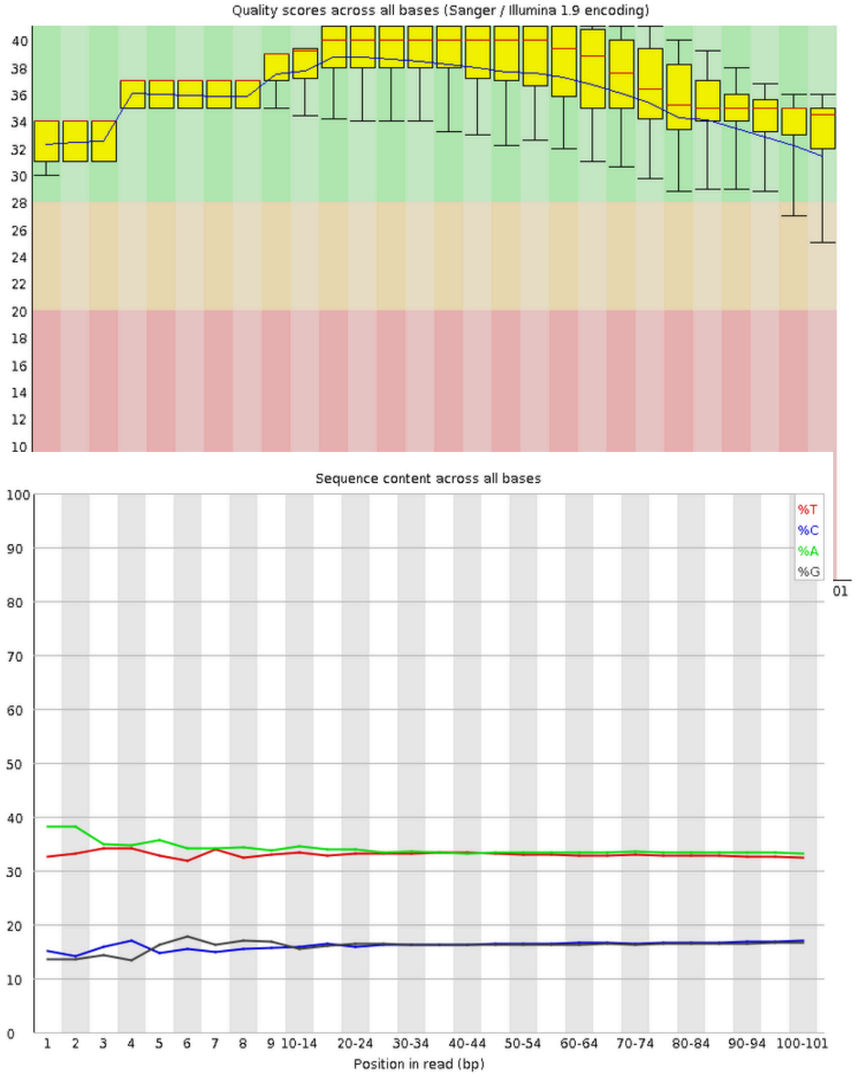
- Remove bad quality reads
- Remove contaminants Trim ends of reads
- Remove orphans (if possible or desirable)
- Correct errors

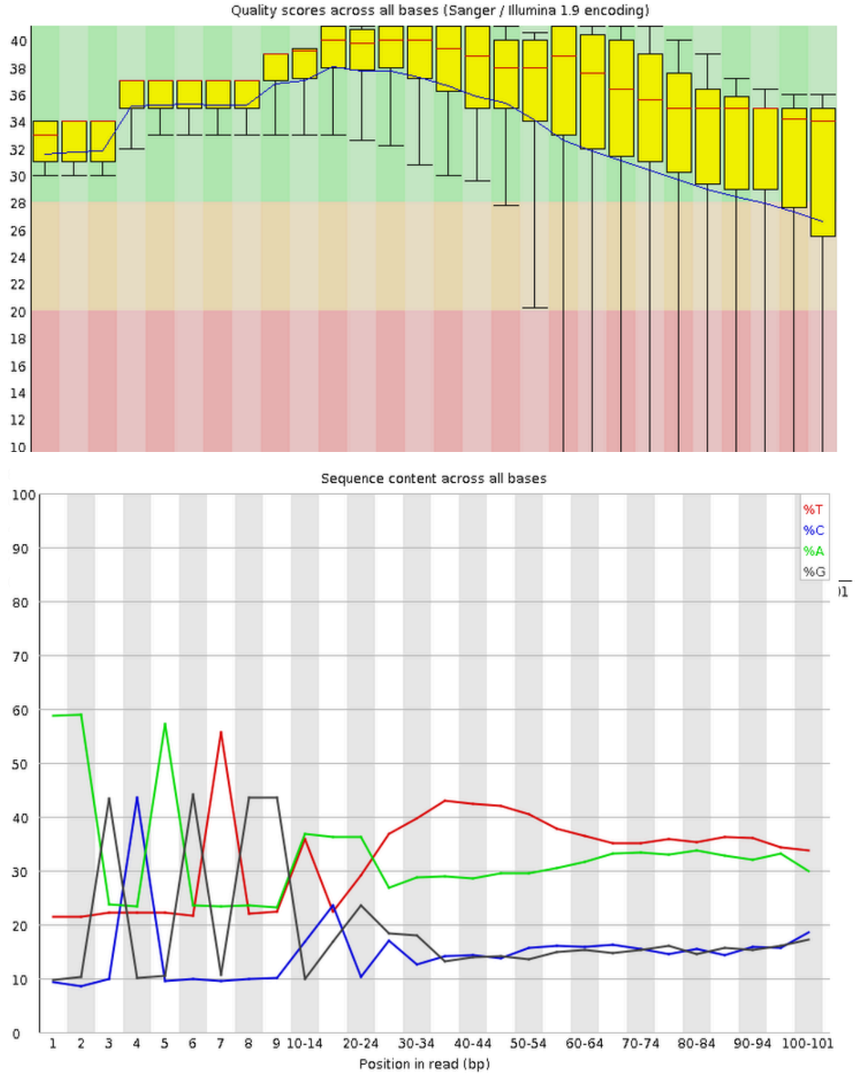


Tools:

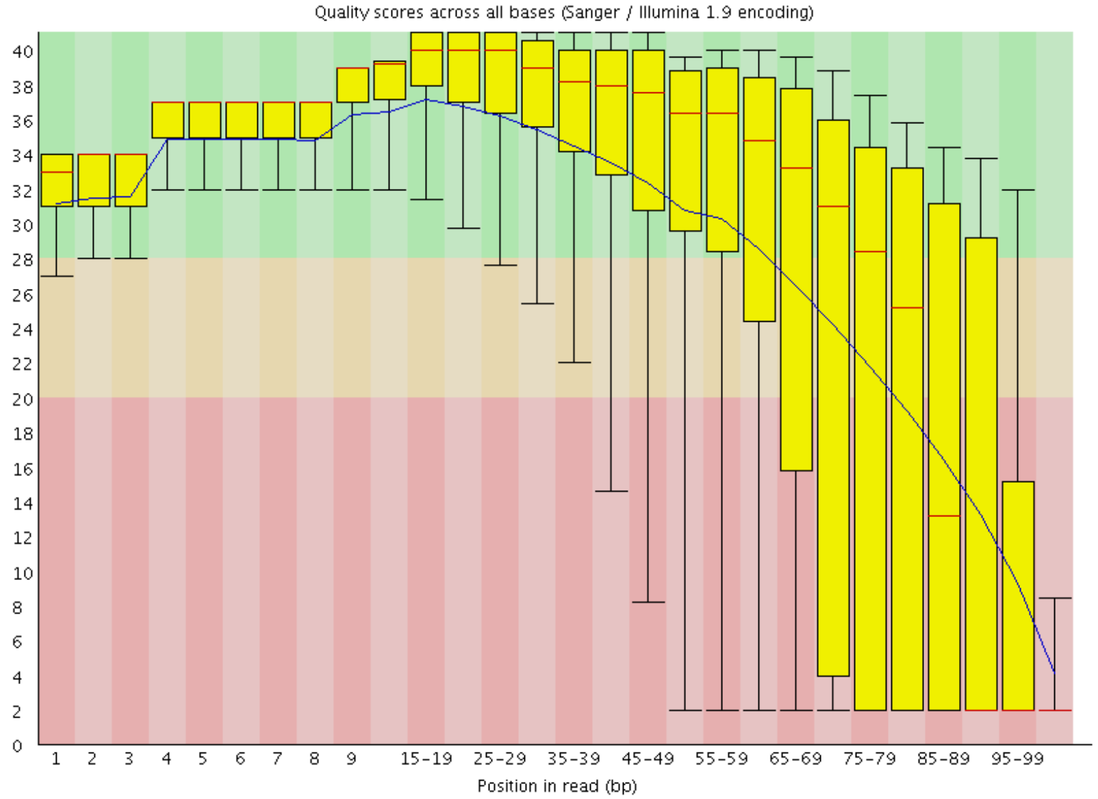
FastQC (http://www.bioinformatics.bbsrc.ac.uk/projects/fastqc/)
MultiQC (http://multiqc.info)
FastX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/)
PrinSeq (http://edwards.sdsu.edu/cgi-bin/prinseq/prinseq.cgi)

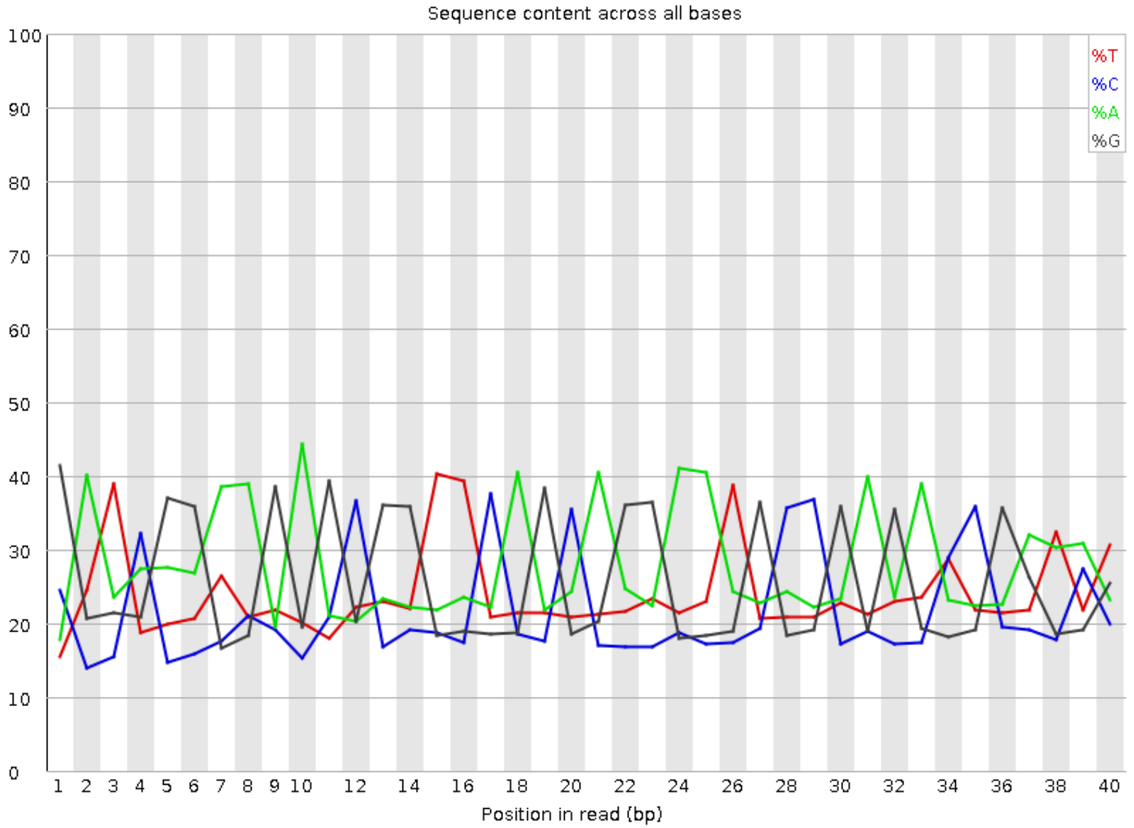
Examples


Quality scores across all bases (Sanger / Illumina 1.9 encoding)
Sequence content across all bases
%T
%C
%A
%G
Position in read (bp)

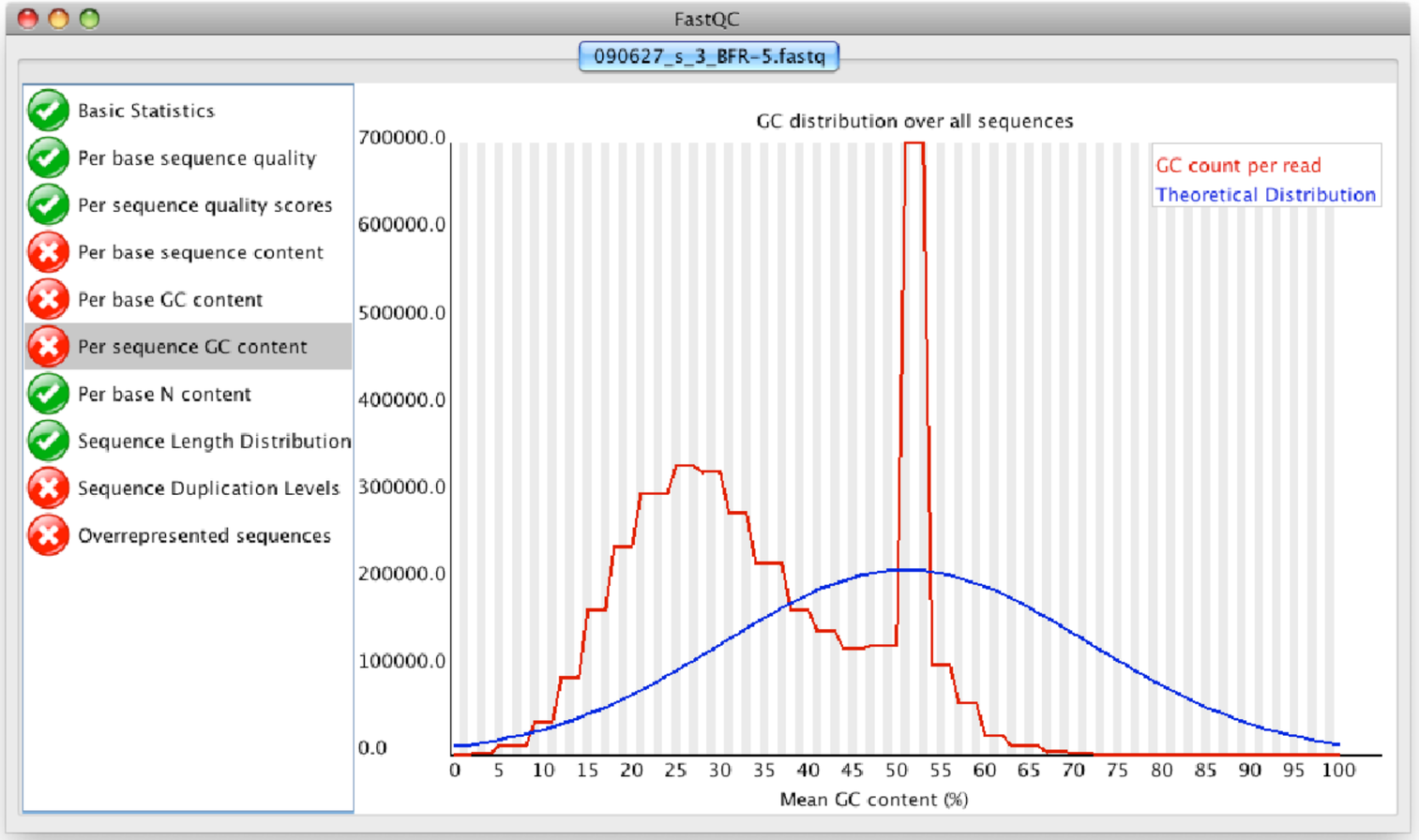

Quality scores across all bases (Sanger / Illumina 1.9 encoding)
Sequence content across all bases
%T
%C
%A
%G
Position in read (bp)

Examples: good or bad?





Examples: contamination



Examples: contamination

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCT	8122	8.122	Illumina Paired End PCR Primer 2 (100% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAG	5086	5.086	Illumina Paired End PCR Primer 2 (97% over 36bp)
AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTAC	1085	1.085	Illumina Single End PCR Primer 1 (100% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGAAG	508	0.508	Illumina Paired End PCR Primer 2 (97% over 36bp)
AATTATACGGCGACCACCGAGATCTACACTCTTTCCCTAC	242	0.242	Illumina Single End PCR Primer 1 (97% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAAGATCGGAA	235	0.23500000000000001	Illumina Paired End Adapter 2 (96% over 31bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCGAGATCGGAAGA	228	0.22799999999999998	Illumina Paired End Adapter 2 (96% over 28bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACG	205	0.20500000000000002	Illumina Paired End PCR Primer 2 (97% over 36bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGATCGGAA	183	0.183	Illumina Paired End Adapter 2 (100% over 32bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGTCGGAAG	183	0.183	Illumina Paired End Adapter 2 (100% over 32bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGAACT	164	0.164	Illumina Paired End PCR Primer 2 (97% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGTCT	129	0.129	Illumina Paired End PCR Primer 2 (97% over 40bp)
AATTATACTTCTACCACCTATATCTACACTCTTTCCCTAC	123	0.123	No Hit
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACT	122	0.122	Illumina Paired End PCR Primer 2 (97% over 36bp)
CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGC	113	0.11299999999999999	Illumina Paired End PCR Primer 2 (96% over 25bp)

Examples: contamination

Overrepresented sequences

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GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCT	8122	8.122	Illumina Paired End PCR Primer 2 (100% over 40bp)
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GATCGGAAGAGCGGTTCAGCAGGAATGCGAGATCGGAAGA	228	0.22799999999999998	Illumina Paired End Adapter 2 (96% over 28bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACG	205	0.20500000000000002	Illumina Paired End PCR Primer 2 (97% over 36bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGATCGGAA	183	0.183	Illumina Paired End Adapter 2 (100% over 32bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGTCGGAAG	183	0.183	Illumina Paired End Adapter 2 (100% over 32bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGAACT	164	0.164	Illumina Paired End PCR Primer 2 (97% over 40bp)
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGTCT	129	0.129	Illumina Paired End PCR Primer 2 (97% over 40bp)
AATTATACTTCTACCACCTATATCTACACTCTTTCCCTAC	123	0.123	No Hit
GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACT	122	0.122	Illumina Paired End PCR Primer 2 (97% over 36bp)
CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGC	113	0.11299999999999999	Illumina Paired End PCR Primer 2 (96% over 25bp)

Other tools: MultiQC

tQC

is a quality control tool for high throughput sequence data, written by Simon Andrews at the Babraham Institute in Cambridge.

ıence Quality Histograms 7 4 1

ın quality value across each base position in the read. See the FastQC help.

Y-Limits

FastQC: Mean Quality Scores



Per Sequence GC Content 2 4 2

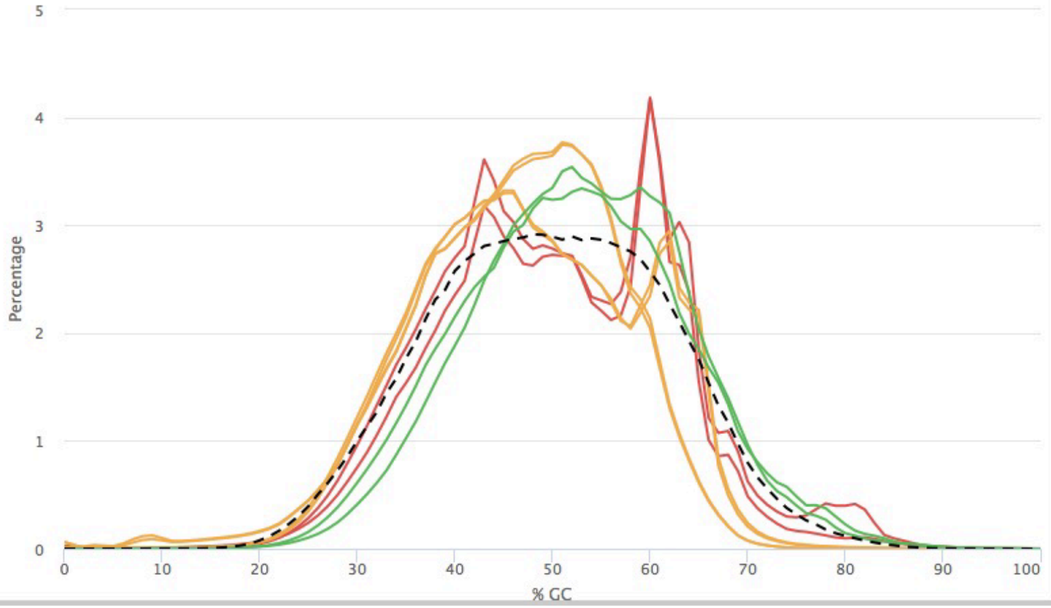
The dashed black line shows theoretical GC content: Human Transcriptome (UCSC hg38).

Y-Limits: on

Percentages | Counts

FastQC: Per Sequence GC Content

Export Plot



Read trimming or filtering

Trimming remove 5' and/or 3' ends of reads (bad quality or adapter)

Filtering remove full reads (e.g., contaminants)

Tools:

- FastX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/)
- PrinSeq (http://edwards.sdsu.edu/cgi-bin/prinseq/prinseq.cgi)
- Sickle (https://github.com/najoshi/sickle)
- ea-utils (https://code.google.com/p/ea-utils/)
- Trimmomatic (http://www.usadellab.org/cms/?page=trimmomatic)
- cutadapt (https://cutadapt.readthedocs.org/)
- Fastp (https://github.com/OpenGene/fastp) and
- Fastplong (https://github.com/OpenGene/fastplong)

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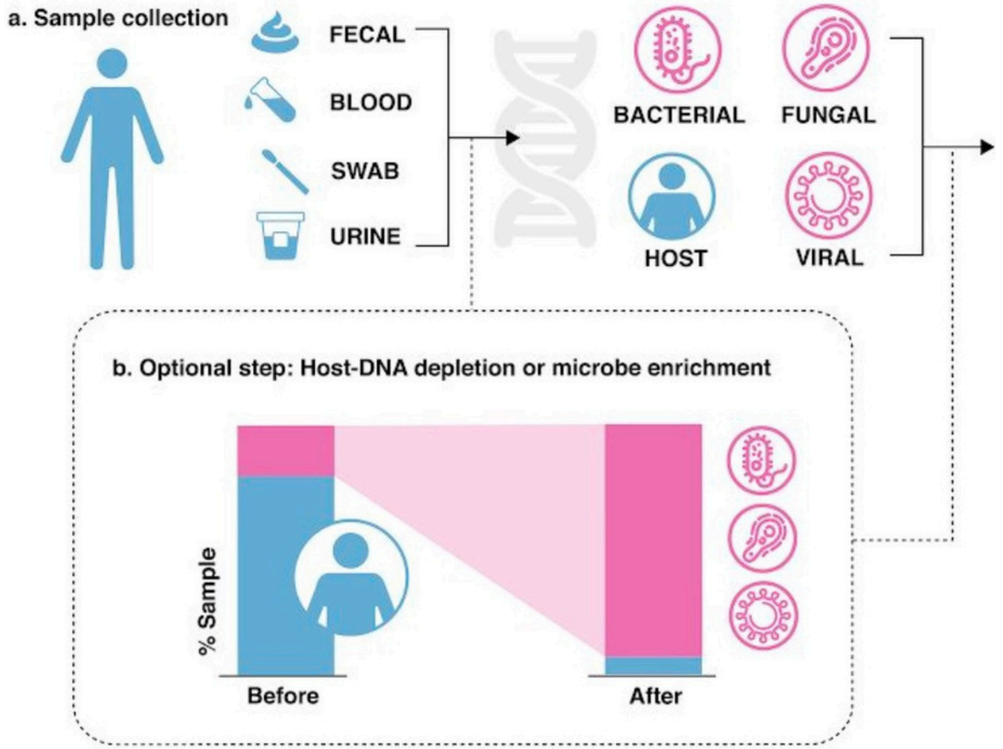
- FastX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/)
- PrinSeq (http://edwards.sdsu.edu/cgi-bin/prinseq/prinseq.cgi)
- Sickle (https://github.com/najoshi/sickle)
- ea-utils (https://code.google.com/p/ea-utils/)
- Trimmomatic (http://www.usadellab.org/cms/?page=trimmomatic)
- cutadapt (https://cutadapt.readthedocs.org/)
- Fastp (https://github.com/OpenGene/fastp) and
- Fastplong (https://github.com/OpenGene/fastplong)

Host removal or depletion

Host DNA sequences are very likely to contaminate your metagenomics data. It is critical to remove those sequences either before sequencing or before analysis.

E.g., PCR amplified 16S RNA can be contaminated by mitochondria or chloroplast 16S of the host.

E.g., WGS can be contaminated by host genomic DNA.



Summary

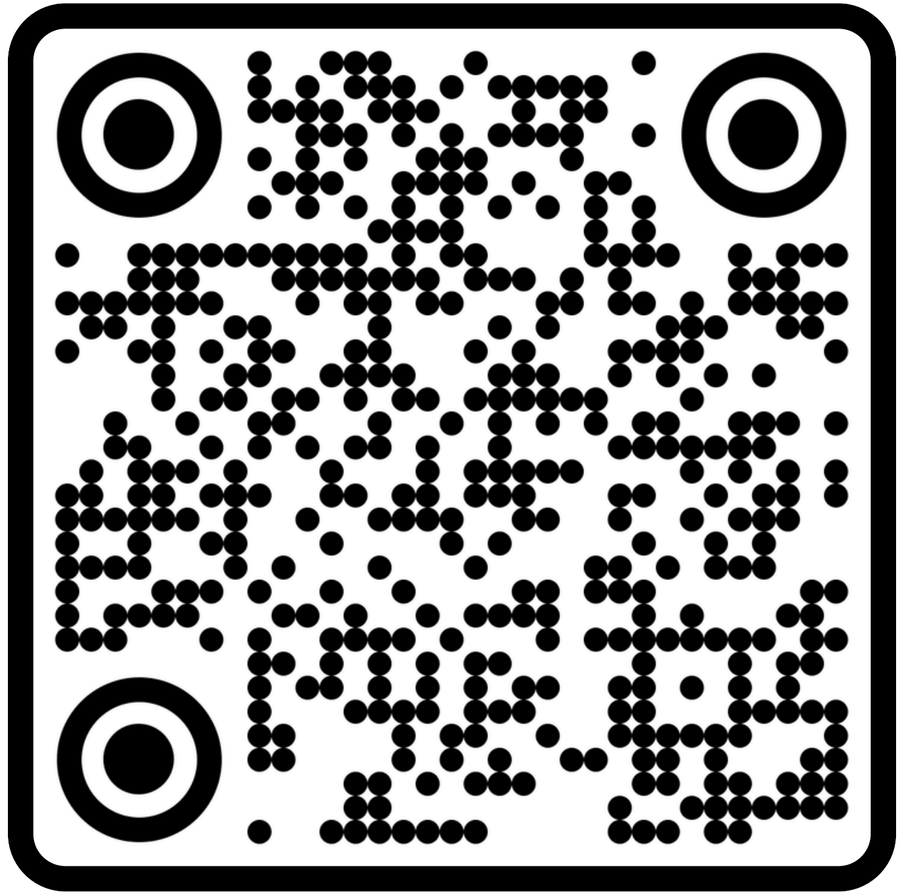
Different sequecing technologies are available

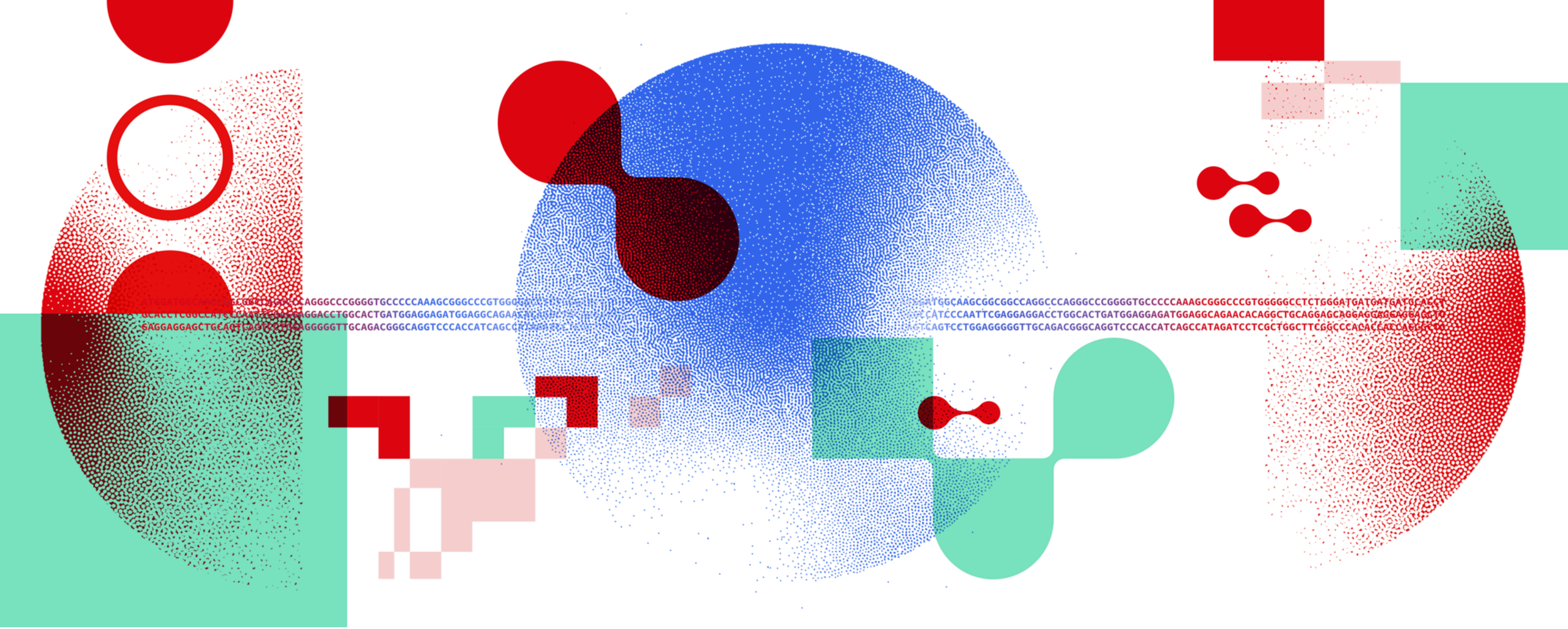
Sequecing technology should be chosen according to the goals of the study

Quality control is necessary to ensure reliability of the results

Analyze carefully and consciously the output of the quality control tools

Practicals





Thank you

elixir SWITZERLAND

SIB